

Multiple Inheritance using Interface

Basic Information

Title: Multiple Inheritance using Interface

File Name: MultipleInheritance_usingInterface.java

Objective / Aim

To demonstrate multiple inheritance in Java using two interfaces implemented by one class.

System Requirements

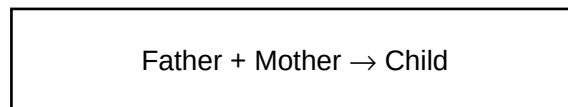
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create Father interface with work().
3. Create Mother interface with cook().
4. Create Child implementing both interfaces.
5. Implement both methods in Child.
6. Create a Child object and call both methods.
7. End the program.

UML Class Diagram



Source Code

```
// First interface
interface Father {

    void work();
}

// Second interface
interface Mother {

    void cook();
}

// Child implements both interfaces
class Child implements Father, Mother {

    public void work() {
        System.out.println("Child is working");
    }

    public void cook() {
        System.out.println("Child is cooking");
    }
}

public class MultipleInheritance_usingInterface {
    public static void main(String[] args) {

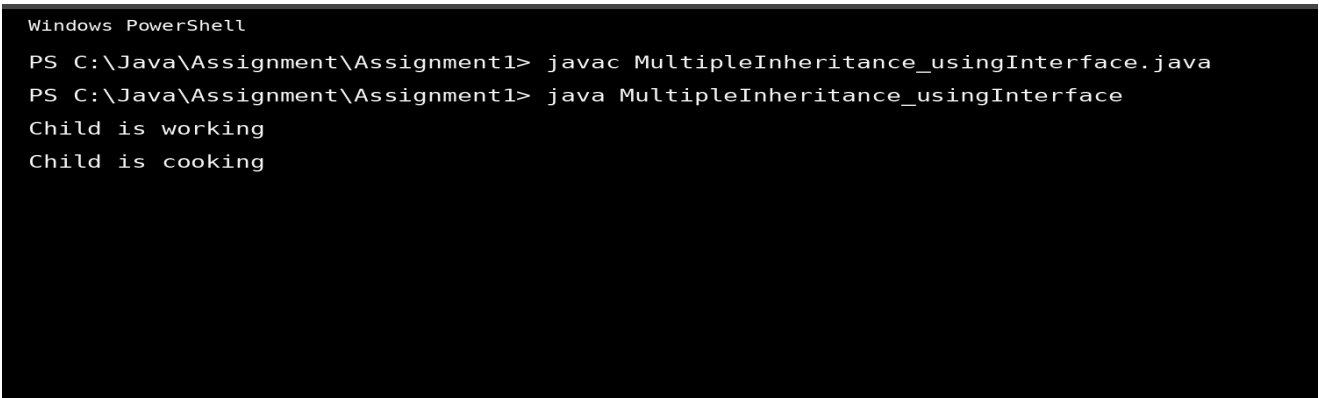
        // Child gets behavior from both interfaces
        Child child = new Child();

        child.work();
        child.cook();
    }
}
```

Compilation & Execution Commands

```
javac MultipleInheritance_usingInterface.java
java MultipleInheritance_usingInterface
```

Output Screen



```
Windows PowerShell
PS C:\Java\Assignment\Assignment1> javac MultipleInheritance_usingInterface.java
PS C:\Java\Assignment\Assignment1> java MultipleInheritance_usingInterface
Child is working
Child is cooking
```

Result

Thus, the Multiple Inheritance using Interface program was implemented successfully using Java inheritance concepts.